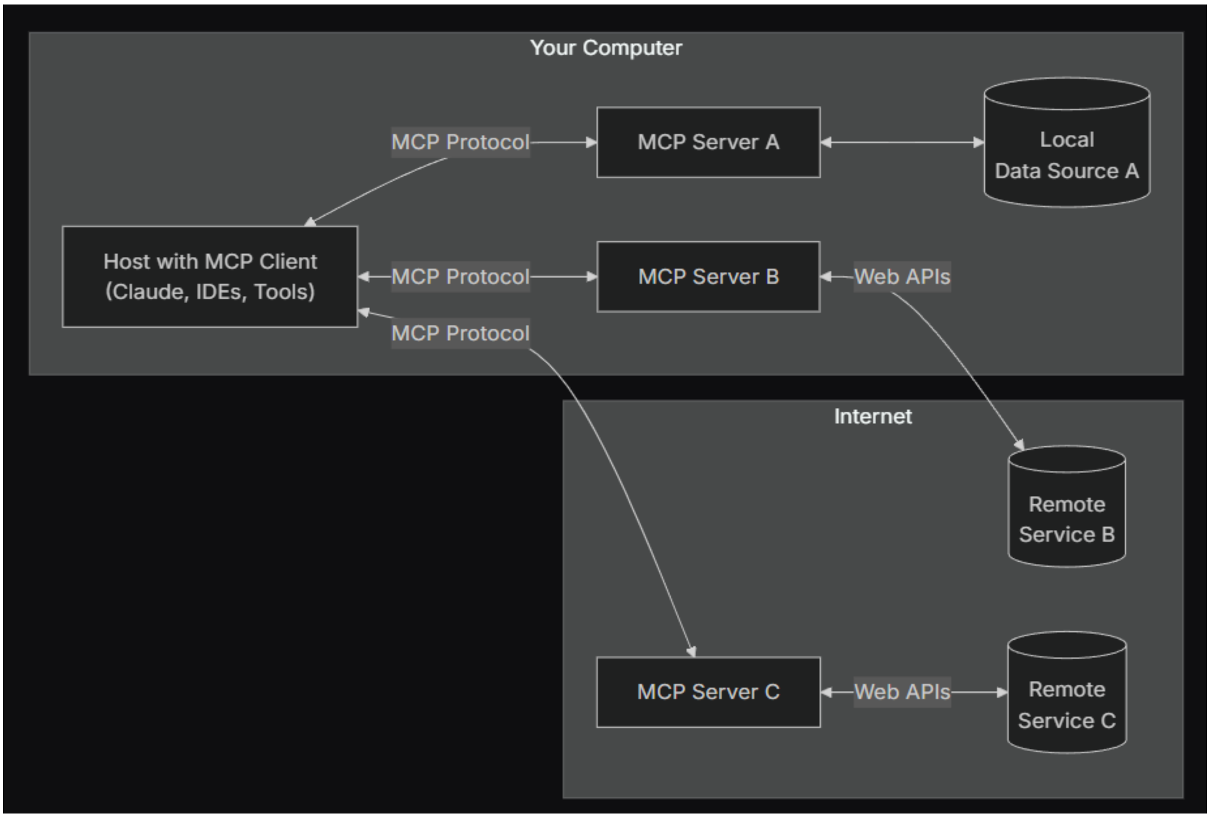
MuleSoft MCP Server

Overview:

The MuleSoft MCP Server is a Model Context Protocol (MCP) implementation that facilitates interaction between large language models (LLMs) and the MuleSoft Anypoint Platform. Create and deploy Mule applications, create API specs, search for assets in Anypoint Exchange, get platform insights, and more.

Simply put, it's a way for AI agents to discover and interact with various tools and resources. An MCP server acts as a bridge, allowing AI models to understand and enable Large Language Models (LLMs) to securely access tools and data sources.

- **Hosts** are LLM applications (like Claude Desktop or IDEs) that initiate connections
- **Clients** maintain 1:1 connection with servers, inside the host application
- **Servers** provide context, tools, and prompts to clients



Pre-requisites:

- Anypoint Extension Pack version 1.9.0 or later.
- Git.
- Organization administrator access.
- Enable Einstein in Access Management.

Implementation Steps:

1. Create a Connected App That Acts on Its Own Behalf in Anypoint Platform.

a) In the Access Management navigation menu, click **Connected Apps**.

b) Optionally, select the business group to create the connected app in.

c) Click **Create App** and enter a unique name for the application.

d) Select **App acts on it's own behalf (client credentials)**.

e) Click **Add Scopes**.

f) Scopes are roles with associated permissions, which determine the actions the app can perform within a business group and environment. For more information, see Endpoint Scopes.

g) In **Add Scopes**, select the scopes to apply and click **Add Scopes**.

h) If applicable, select the business groups to apply the scopes to and click **Next**.

i) If applicable, select the environments to apply the scopes to and click **Next**.

j) Click **Add Scopes**, and then click **Save**.

2. Add these scopes.

Make sure that all business groups and relevant environments are selected for these scopes.

a) **Anypoint Code Builder -** Mule Developer Generative AI User
b) **Anypoint Monitoring -** Monitoring Viewer
c) **API Manager -** Manage APIs Configuration, Manage Policies and View Policies
d) **Exchange -** Exchange Administrator, Exchange Creator and Exchange Viewer
e) **General -** View Organization and View Connected Applications
f) **Runtime Manager -** Read and Create Applications, Read Runtime Fabrics and Cloudhub Network Viewer.
g) **Usage -** Usage Viewer

3. Install the MuleSoft MCP Server

To install the server with node, run: npm install -g @mulesoft/mcp-server

4. Configure the MuleSoft MCP Server with host

You can use below configuration for configure Claude Desktop, Zed, Cursor, Windsurf, and other IDEs to work with the MuleSoft MCP Server.

```
{
  "mcpServers": {
    "mulesoft": {
      "command": "npx",
      "args": ["-y", "@mulesoft/mcp-server", "start"],
      "env": {
        "ANYPOINT_CLIENT_ID": "<YOUR_CLIENT_ID>",
        "ANYPOINT_CLIENT_SECRET": "<YOUR_CLIENT_SECRET>",
        "ANYPOINT_REGION": "<REGION_NAME>"
      }
    }
  }
}
```

Available Tools:

Application Development:

- **create_and_manage_assets:** Create an asset in Anypoint Exchange, update some Anypoint Exchange asset properties without republishing the asset, and delete an asset in Anypoint Exchange. Asset properties are the asset name, description, icon, contact name, and contact email.
- **create_api_spec:** Create an API spec using AI. Prerequisites: Log in to Anypoint Code Builder with your Anypoint Platform credentials. Make sure Einstein is enabled in Access Management.
- **create_api_spec_project:** Create and scaffold an API spec or fragment project. You can specify the project name, location, and spec language (RAML, OAS, etc.).
- **create_MCP_server:** Create a MuleSoft MCP server. This tool handles requests for creating MCP servers using the Anypoint Connector for MCP (MCP Connector). Prerequisite: Install the MCP Connector.
- **create_mule_project:** Create an integration project.
- **generate_mule_flow:** Create a flow using AI.
- **run_local_mule_application:** Deploy and run your application locally.
- **search_asset:** Search for an asset in an organization and Anypoint Exchange. Returns the first asset in the user's own organization that matches input criteria. If no assets are found in the user's organization, then the tool returns assets in Anypoint Exchange that exactly or partially match search Query.

Application Management:

- **create_and_manage_api_instances:** Create and configure API instances in Anypoint API Manager.
- **deploy_mule_application:** Deploy your application to CloudHub 2.0 or Runtime Fabric and run it. You can deploy from a local project or Anypoint Exchange asset. This tool supports security, high availability, and performance optimization deployment options.
- **list_api_instances:** Get information for instances in an environment. You can also filter by an instance ID or label and include applied or available policies. Results are paginated using the page parameter. Prompt tip: To get information for all instances, revise results using the page parameter.
- **list_applications:** Get information about all applications in the environment you specify. You can also specify an optional deployment name. Results are sorted by last update date and time, in descending order.
- **update_mule_application:** Update a Mule application deployed to CloudHub 2.0 or Runtime Fabric. Update runtime version, configuration, and other application settings.

Policy Management:

- **manage_api_instance_policy:** Apply a new policy to an API instance and update an existing policy configuration.
- **manage_flex_gateway_policy_project:** Provide instructions to the LLM on how to manage a Flex Gateway custom policy project. Instructions include how to create the project, set up the configuration, build the project, publish a development version of the project, and publish a release version of the project.
  Prerequisites: https://docs.mulesoft.com/pdk/latest/policies-pdk-prerequisites
- **get_flex_gateway_policy_example:** Get a Flex Gateway policy example to help you configure your Rust source code. You can get example policies that manipulate a request's header and body, make HTTP calls to external services, share data between requests and responses, reject requests, and perform sleep operations.

Usage & Insights:

- **get_platform_insights:** See how your organization leverages the Anypoint Platform through usage and operational insights. These include:
  - Usage trends for apps, such as flow count, message counts, and data throughput by business group and environment. Usage trends are for the current month. If it's early in the month, you get usage for the last full month.
  - API and app call volume across consumers for last 7 days.
  - API and app performance metrics such as errors and latency for last 7 days.
- **get_reuse_metrics:** Get visibility into how assets are reused across your organization. Drive efficiency, consistency, and cost savings by reducing duplication. Includes sandbox and production environments. Metrics are a year-to-date snapshot.

Example Commands:

If you want to…	Try this…
Apply a policy	Apply the rate-limiting policy to this API instance.
Create an asset in Exchange	I want to publish this API to Exchange.
Create an instance	Create an instance for this application.
Create an integration project	Create a Mule project that creates an order in NetSuite every time an opportunity in Salesforce is updated to stage Closed Won.
Deploy an application	Deploy Mule application in current project. Deploy with high security, high availability, and performance optimized settings.
Deploy an existing asset in Anypoint Exchange to CloudHub 2.0 or Runtime Fabric	Deploy app using sample-asset-name from Anypoint Exchange.
Generate a MCP server using Anypoint connectors	Generate an MCP server that exposes 3 tools from the following operations: Workday - Absence Management - Get Time Off Plan Balances, Workday - Absence Management - Enter Time Off, and Google Calendar - Create Calendar Event.
List all policies	What policies do I have available?